Chemistry 3A – Spring 2026 «First» «Last» «Student_ID» RETAKE VERSION

Sections 19320-19321

Midterm Examination #1 «Exam_Date»

Indicate **BY DIRECTLY CIRCLING** CLEARLY on the exam page *one* and *only one* choice that is **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Which element belongs to the **alkali metals** group?
   (a) calcium (b) sodium (c) magnesium (d) chlorine

2. Which subatomic particle has a mass of approximately **1 amu** and a positive charge?
   (a) electron (b) proton (c) neutron (d) none of the above

3. Which of the following is a **polyatomic ion**?
   (a) Cl^- (b) Na^+ (c) SO_4^{2-} (d) O^{2-}

4. Convert 25.0 °C into Kelvin.
   (a) 0 K (b) 25 K (c) 273 K (d) 298 K

5. A sample has a mass of 12.5 g and a volume of 2.50 mL. What is its density?
   (a) 0.200 g/mL (b) 5.00 g/mL (c) 31.3 g/mL (d) 0.050 g/mL

6. Which of the following is the correct name for $Ca(NO_3)_2$?
   (a) cobalt fluoride (b) calcium nitroxide (c) calcium nitrate (d) calcium oxide

7. Which of the following terms describes the distance between two wave crests?
   (a) wave speed (b) frequency (c) wavelength (d) amplitude

8. Which of the following is the correct atomic notation for carbon-14?
   (a) ${}^{12}_{6}C$ (b) ${}^{14}_{6}C$ (c) ${}_{14}^{6}C$ (d) ${}^{60}_{27}Co$

9. Convert 4.56×10^{-3} into standard notation
   (a) 0.00456 (b) 4560 (c) 0.000456 (d) 4,560

10. Which of the following is the electron configuration for hydrogen?
    (a) $1s^1$ (b) $1s^2$ (c) $1s^2 2s^1$ (d) $1s^2 2p^1$

11. Which of the following is TRUE about element bromine (Br)?
    (a) It is a member of the *d*-block of elements
    (b) It is a Group 1 element
    (c) It is a non-metal
    (d) none of the above

12. What term describes an electron normally in $n = 1$ shell acquiring energy to be in $n=2$ shell?
    (a) emission (b) excited state (d) ground state (d) nucleated

13. Express 0.0000721 in scientific notation with correct significant figures.
    (a) 7.21×10^{-4} (b) 7.21×10^{-5} (c) 7.21×10^{-6} (d) 7.21×10^{-7}

14. When you complete the operation on the expression $1.450 \times 3.01 \div 10.025$, how many significant digits will the final result have?
    (a) 2 (b) 3 (c) 4 (d) 5 (e) 7

15. The scientific method includes a stage where a **tentative explanation** is given to account for a set of related observations. What is that stage?
(a) law (b) hypothesis (c) theory (d) experimentation

16. Which of these is a polyatomic ion?
(a) Ne (b) CO_3^{2-} (c) Br^- (d) Na^+

17. Which of these is the ground-state electron configuration for the **carbon (C)** atom?
(a) $1s^2 2s^2 2p^6$ (b) $1s^2 2s^2 2p^2$ (c) $1s^2 2s^2 2p^4$ (d) $1s^2 2s^2$

18. Electrons have a set of quantum numbers that give them an identity or uniqueness in the atom. How many quantum numbers can they have?
(a) 1 (b) 2 (c) 3 (d) 4

19. Perform the operation: 3.456 + 12.1 + 0.0045. Report with correct decimal places.
(a) 15.5605 (b) 15.56 (c) 15.6 (d) 15.561

20. The label on the bottle has the formula $FeCl_3$. What is its name using the Stock system?
(a) iron chloride (b) iron(II) chloride (c) iron(III) chloride (d) iron monochloride

21. One of the quantum numbers of an electron is related to the shape of the electron's orbital and is represented by a letter in this set: {*s*, *p*, *d*, *f*}. Which quantum number is this?
(a) *n* (b) *l* (c) m_s (d) Z

22. Which of these subatomic particles is **NOT** found in the nucleus of the atom?
(a) proton (b) neutron (c) electron (d) all the choices are NOT in the nucleus

23. What is a TRUE statement regarding an element in the Periodic Table?
(a) the number of neutrons and protons for every element must be equal
(b) the atomic number (Z value) indicates the number of protons it has
(c) the mass number (A value) indicates the number of electrons and protons it has
(d) the isotope of the element is determined by the number of electrons

24. You have a volume of 650.0 mL. One and only one of the values below is equivalent and another way of expressing that value. Which one is it?
(a) 6.500×10^3 L (b) 0.650 L (c) 6.5 L (d) 0.0006500 mL

25. What is the atomic number of zinc (Zn)?
(a) 65.39 (b) 29 (c) 30 (d) none of the choices (a) to (c)

Circle either TRUE or FALSE whether the statement is true or false.

26. A **cation** is an ion that has a net negative charge because it gained one or more electrons.
True False

27. Ionic compounds with monatomic elements are typically formed between metals and nonmetals.
True False

28. The speed/velocity of a electromagnetic wave can be calculated by dividing the frequency by the wavelength: $c = \nu / \lambda$
True False

29. The **p-block** of elements in the Periodic Table starts in Period 1.
True False

30. The **Pauli Exclusion Principle** states that an orbital can have only two electrons which are paired up with opposite spins
True False

31. According to the rules for significant figures, **conversion factors** (like 1 m = 100 cm) are considered to have infinite significant digits and do not affect the precision of a calculation.
True False

32. The atomic number (*Z*) represents the number of protons in an atom.
True False

33. The only subatomic particles allowed in the nucleus of the atom are **protons**.
True False

34. Noble gases have complete valence shells, which is why they do not form compounds with other elements.
True False

35. The law of conservation of mass states that matter **CAN** be created or destroyed in chemical reactions.
True False